

3.6 Alternating Series Worksheet

Determine whether each of the following series converge or diverge. You may use any technique from 3.6 or earlier.

1.
$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln(n+4)}$$

2.
$$\sum_{k=1}^{\infty} \frac{(-3)^{k+1}}{2^k}$$

3.
$$\sum_{n=1}^{\infty} \frac{2n}{n^2+1}$$

4. $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2 + 2}$

5. $\sum_{n=1}^{\infty} (-1)^n e^{2/n}$

6. $-\frac{1}{1} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} - \frac{1}{5^2} + \frac{1}{6^2} - \dots$ where there are always two negative terms followed by a positive term.

7. The series $\sum_{n=1}^{\infty} \left(e^{-n^2} - e^{-(n+1)^2} \right)$ converges. Determine what value the series converges to.

8.
$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2n+1}{3n-1}$$

Classify each of the following series as absolutely convergent, conditionally convergent, or divergent.

9. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n+1}$

10. $\sum_{k=1}^{\infty} \frac{\cos(k)}{k^2}$

$$11. \sum_{k=1}^{\infty} \frac{(-1)^{k+1} k}{2k^3 + 1}$$

$$12. \sum_{n=1}^{\infty} \frac{(-1)^n (n+3)}{n}$$

$$13. \sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n+3}}{n}$$

14. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^6}$ converges by the Alternating Series Test.

a) Determine the upper bound on how far off the 5th partial sum of the series is from the actual sum of the series.

b) How many terms of the series are needed in order to find the sum of the series to within 0.00005.

15. The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n \cdot 5^n}$ converges by the Alternating Series Test.

a) Determine the upper bound on how far off the 2nd partial sum of the series is from the actual sum of the series.

b) How many terms of the series are needed in order to find the sum of the series to within 0.0001.

16. The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^4}$ converges by the Alternating Series Test.

a) Determine an upper bound on how far off the 10th partial sum of the series is from the actual sum of the series.

b) What is the first partial sum of this series that we can guarantee is within one thousandth of the actual sum?

17. The series $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n+3}}{n}$ converges by the Alternating series test.

a) If this sum is estimated by the 21st partial sum, determine an upper bound for the error on this approximation.

b) What is the first partial sum of this series that we can guarantee is within one tenth of the actual sum?